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Melting behaviour and homogeneity range of *B*2 CoAl and updated thermodynamic description of the Al–Co system

F. Stein [a,*], C. He [b], N. Dupin [c]

[a] Max-Planck-Institut für Eisenforschung GmbH, Max-Planck-Strasse 1, D-40237 Düsseldorf, Germany
[b] College of Material Science and Engineering, Guangxi University, Nanning 530005, Guangxi, PR China
[c] Calcul Thermodynamique, F-63670 Orcet, France

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ABSTRACT

The central part of the Al–Co phase diagram is dominated by the high-melting *B*2 CoAl phase. Although this phase is well-known since long and has been intensively investigated in the literature, neither its melting behaviour nor its homogeneity range is well established. Therefore, a series of Al–Co alloys in the composition range 30–60 at.% Co were produced and analysed by differential thermal analysis, scanning electron microscopy, and electron-probe microanalysis. The composition dependence of the melting temperatures was determined and from the analysis of heat-treated, two-phase CoAl + Al_5Co_2 samples, the Al-rich phase boundary of the CoAl phase and the Co-rich boundary of the Al_5Co_2 phase were established. The melting temperatures of CoAl were found to be significantly higher than previously reported in the literature; for the stoichiometric compound a value of 1673 °C was obtained. On the basis of these new results and further relevant literature data, the complete Al–Co phase diagram was finally thermodynamically re-assessed applying the CALPHAD technique. The optimized thermodynamic parameters not only give a good description of the new experimental data, but also fit very well to most of the literature data.

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1. Introduction

Aluminides with an ordered *B*2-type crystal structure such as FeAl, CoAl and NiAl belong to the most intensively studied intermetallic compounds. On the one hand they are regarded as attractive candidates for the development of high-temperature materials for structural applications due to their (compared to steels) low densities, excellent oxidation behaviour and (except FeAl) high melting temperatures. On the other hand they are interesting materials from the purely scientific point of view. Their comparably simple crystal structure and extended homogeneity ranges make them well suited for studying basic properties of intermetallic phases as for example the role of point defects [1,2].

The Al–Co phase diagram was studied for the first time by Guillet as early as in 1902 [3,4], and only six years later, a complete phase diagram was published by Gwyer [5] resulting from analyses of time-vs.-temperature cooling curves and metallographic investigations of about 30 different alloys. He identified three intermetallic phases which according to their compositions were named CoAl, Co_2Al_5 and Co_3Al_{13}. For a nearly stoichiometric CoAl alloy with a composition of 49.9 at.% Co, he obtained a liquidus temperature of 1628 °C. The same value for the maximum melting temperature of CoAl was reported in 1933 in the text of a paper by Köster [6] even though a figure in his article, which shows the partial phase diagram 51.7–100 at.% Co, indicates a higher liquidus temperature of approximately 1650 °C. However, five years later he gives the maximum temperature as 1620 °C [7]. In 1974, Panteleimonov et al. [8] presented a complete phase diagram of the binary system indicating a melting temperature of 1645 °C for CoAl. However, it is not clear whether this value results from a measurement or is extrapolated from other experimental data. Since then, a great many of further studies on the Al–Co system has been published. However, either they focus on the Al-rich part of the phase diagram, where a series of peritectically forming intermetallic phases occurs being a subject of some controversy [9–12], or they deal with the Co-rich part, where the (Co) solid solution forms a two-phase field with the *B*2 CoAl phase. Thorough assessments of all existing data on the Co–Al phase diagram have been published by McAlister [13] and by Dupin and Ansara [14]. The latter paper additionally took into account experimental thermodynamic data and derived a Calphad description. Fig. 1 shows the Al–Co phase diagram as given in [13,15]. On the basis of the data of Gwyer and

* Corresponding author. Tel.: +49 211 6792 557; fax: +49 211 6792 299.
E-mail address: stein@mpie.de (F. Stein).

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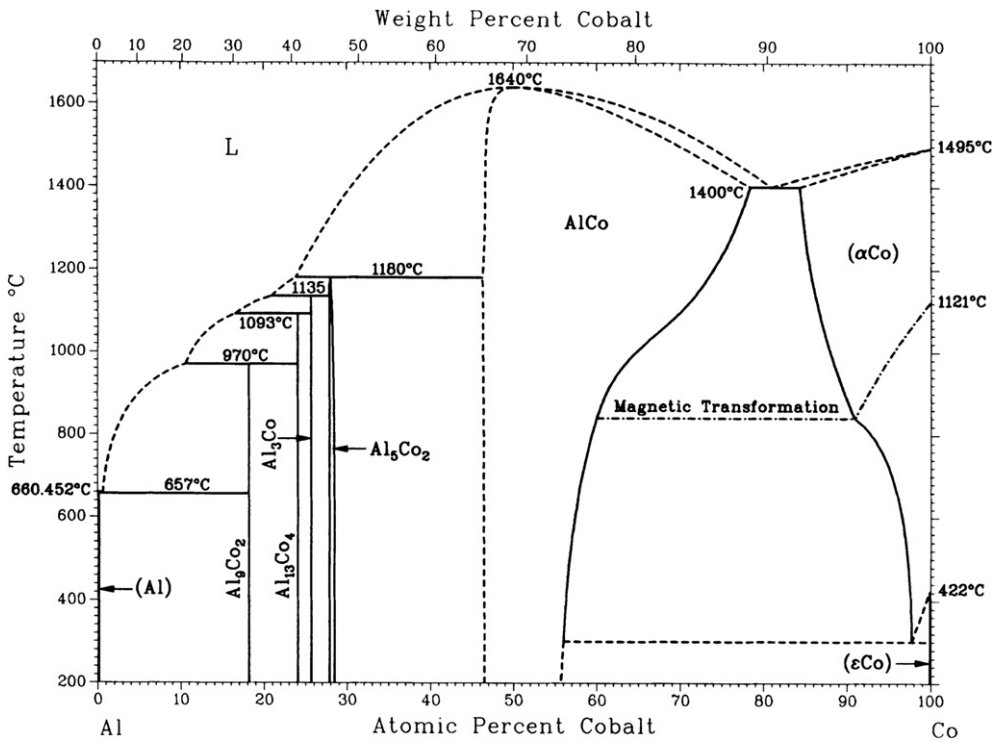

Fig. 1. Previously accepted Al–Co phase diagram as shown in [13,15].

Köster and "in the absence of better information", McAlister [13] drew the liquidus line in the central part of the system with a maximum at 1640 °C, see Fig. 1.

Even though McAlister clearly states that for the *B*2 CoAl phase "there is reason to distrust available data concerning its melting behaviour" and, therefore, draws the liquidus as broken line, the value of 1640 °C for the melting point of stoichiometric CoAl is still used until today in all publications dealing with *B*2 CoAl. No other experimental data on the melting point of CoAl than those of Gwyer [5], Köster [6,7], and Panteleimonov et al. [8] were reported since then. Calculations by *ab initio* methods yielded significantly varying, more or less rough estimates of the melting temperature (Mehl et al. [16]: ~1800 °C, del Grosso et al. [17]: 1592 °C, Dong et al. [18]: 1627 °C). The only additional experimental information available from literature for the melting curve of *B*2 CoAl is for the Co-rich composition region near the CoAl + (αCo) eutectic [19,20]: By thermal analysis, Schramm [19] determined some liquidus temperatures for alloys with 72 and more at.% Co. Compared to Gwyer's results the values are about 50–60 °C higher. In a much more recent investigation, Kimura et al. [20] measured solidus and liquidus temperatures of alloys with 64 and more at.% Co by differential thermal analysis (DTA). Although to a less extent, their values, too, are clearly higher than those of Gwyer. McAlister speculates in his critical assessment [13] that these differences can be attributed to problems with material purity in case of Gwyer's data.

As the phase diagram in Fig. 1 shows, not only the liquidus line is drawn with broken lines but also the exact position of the Al-rich phase boundary of CoAl is unknown. The high solubility of Co in the *B*2 CoAl phase at high temperatures strongly decreases with temperature. This is well documented by various experimental studies summarized in [13,14] and leads to the interesting effect of the formation of metastable *A*2 + *B*2 two-phase microstructures (instead of the expected *A*1 + *B*2 equilibrium microstructure) in Co-rich alloys quenched from the *B*2 CoAl single-phase field as was found experimentally [21–23] and theoretically [24]. In contrast to the Co-rich boundary, nearly no information is available on the position of the Al-rich boundary. The activity of aluminium in solid Al–Co alloys, which had been measured by an isopiestic technique and presented as a function of composition at 1000 °C [25], shows a pronounced decrease at the Al-rich boundary of the single-phase field. From this plot, McAlister [13] has extrapolated a value of 46.4 at.% Co for the boundary. The only other information available on this phase field boundary is from early XRD investigations of Ekman [26], who reported an alloy with 47.3 at.% Co to be single-phase CoAl after it had been heat-treated for one week at 880 °C.

In the following, a re-investigation of the central part of the Al–Co phase diagram is described. Alloys in the composition range 30–60 at.% Co were studied by DTA to obtain the melting temperatures of CoAl (preliminary experimental data on the melting temperature of CoAl have already been used in a recent publication on the liquidus surface of the Al–Co–Nb system [27]) and selected heat-treated samples were analysed by electron-probe microanalysis (EPMA) to fix the Al-rich boundary of the CoAl single-phase field. Finally, using the Calphad method, the complete Al–Co phase diagram was re-assessed by taking into account the new experimental data and relevant literature data. As a starting point for the present assessment, the description of Dupin and Ansara [14] was used.

2. Experimental work

2.1. Experimental

From the pure metals Al (99.999 wt.%) and Co (99.95 wt.%), eight binary alloys in the composition range 30–60 at.% Co were prepared by conventional melting procedures under an argon atmosphere. The nominal and analysed compositions of all investigated alloys are given in Table 1. For further examinations, the alloys were sectioned by electro-discharge machining (EDM). Wet chemical analyses of two as-cast samples yielded impurity contents in the

Table 1
Nominal and analysed alloy compositions of the as-cast Al–Co alloys, preparation method (IM: induction melting, AM: arc melting), and solidus temperatures T_{sol}, liquidus temperatures T_{liq}, and enthalpy of melting ΔH_{melt} as determined by DTA. The accuracy of the enthalpy values is ±20%.

Alloy composition (at.% Co)		Preparation method	T_{sol} (°C)	T_{liq} (°C)	ΔH (J/g)
Nominal	Analysed by EPMA				
30	30.5	IM	1180 ± 1	1395 ± 10	
40	40.7	IM	1181 ± 1	1618 ± 5	
48	48.6	IM	1629 ± 5	1670 ± 5	590
49	49.8	IM	1673 ± 3	1673 ± 4	590
50	50.4	IM	1673 ± 3	1676 ± 5	520
55	55.6	IM	1598 ± 5	1656 ± 5	460
57	57.2	IM	1565 ± 10	1645 ± 4	430
60	59.4	AM	1525 ± 5	1625 ± 7	280

range 50–80 wt. ppm oxygen, 10 wt. ppm nitrogen and 20 to 30 wt. ppm carbon.

For heat treatments, pieces of the alloy Al–40.7Co (all compositions given in at.% throughout this paper) were encapsulated in quartz ampoules which were evacuated and back-filled with argon several times before sealing them. After heat treatments for 3 weeks at 900 °C or 1 week at 1100 °C, the samples were quenched by breaking the ampoules in iced brine (10% NaCl solution).

For metallographic observation of the microstructures, light-optical (LOM) and scanning electron microscopy (SEM) were used. For LOM investigations, the samples were etched by a solution of glycerine (68 vol.%), 70%-HNO_3 (16 vol.%), and 40%-HF (16 vol.%). SEM was performed using the EPMA instrument described below in SEM mode.

DTA experiments were carried out with a Setaram SETSYS-18 DTA instrument (SETARAM, Caluire, France) allowing measurements up to 1750 °C. Cylindrical samples of 3 mm in diameter and height were placed in alumina crucibles and heated and cooled at least twice with a rate of 5 K/min (and in particular cases additionally with 2 K/min) in an Ar atmosphere. Calibration measurements were performed using certified standards of pure Au, Ni, and Pd resulting in an accuracy of ±1 K for the measured temperatures. The values for the enthalpy of melting, which were determined from the area below the melting peaks, have a much higher relative error which is tentatively estimated to be ±20%.

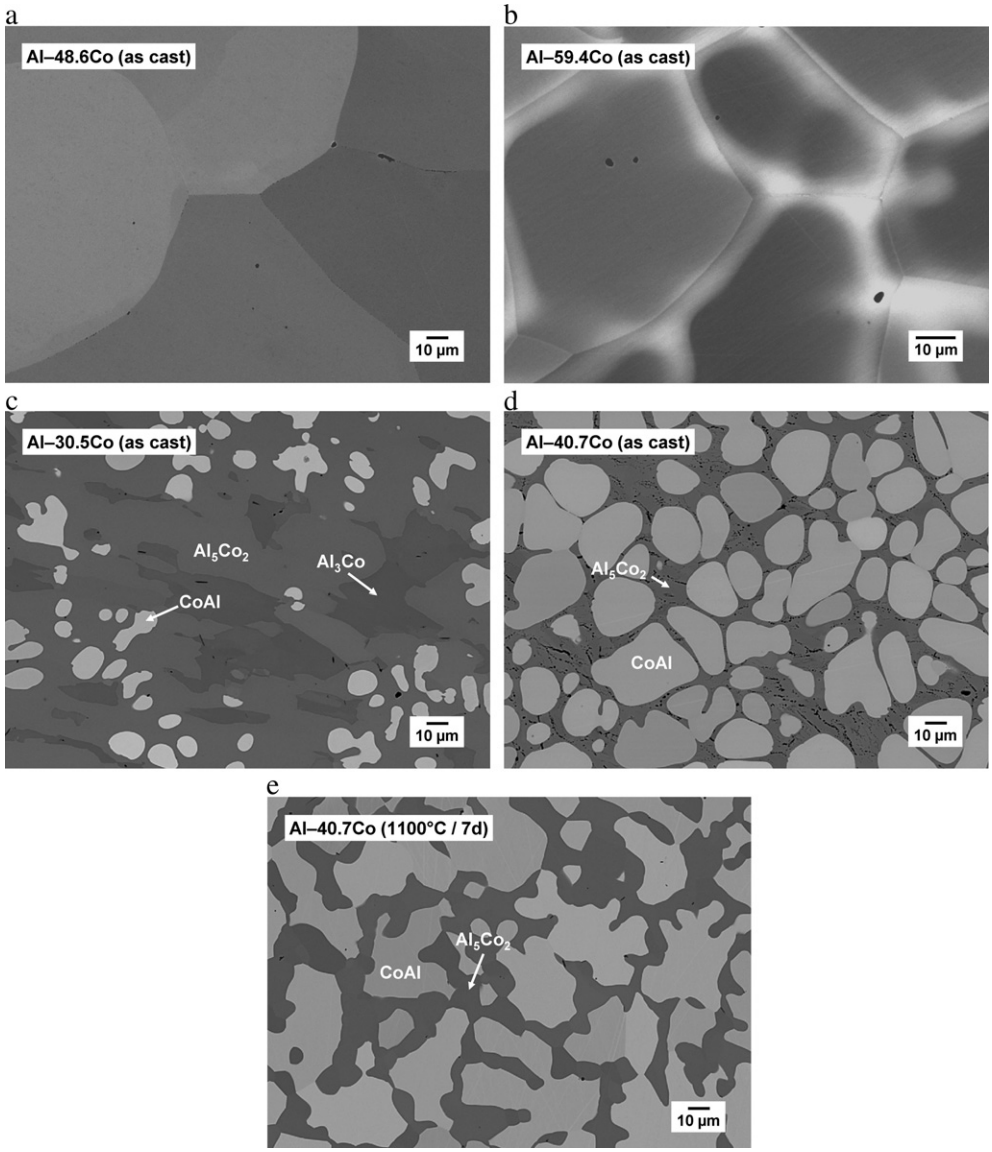

Fig. 2. SEM micrographs (back-scattered electron contrast) showing the microstructure of selected as-cast (a–d) and heat-treated (e) Al–Co alloys.

EPMA measurements were carried out with a Jeol JXA-8100 instrument using pure Co and Al as standards. Phase compositions were analysed with a beam probing sample volumes of about 1 μm^3 and averaging at least twelve measured points per phase. The relative error of the resulting compositions is 1%. The actual bulk compositions of the alloys were determined by averaging the EPMA results of 750–800 spot analyses per sample with spacings of 10 μm in three rectangular fields of about 100–300 μm in width and length. The EPMA values were finally calibrated by a comparison with values obtained by inductively coupled plasma optical emission spectroscopy (ICP-OES).

2.2. As-cast microstructures and phases

Representative microstructures of as-cast alloys are shown in Fig. 2. The investigated alloys with 48.6–59.4 at.% Co are single-phase consisting of the *B*2 phase CoAl (Fig. 2a,b). The nearly stoichiometric CoAl alloys have a very homogeneous microstructure, the standard deviations of about 800 measured EPMA composition values are below 0.4 at.%. In contrast to that, the Co-rich single-phase alloys reveal strong composition gradients, which result from the split of the liquidus and solidus curves in the range between 50 and 80 at.% Co (see phase diagram in Fig. 1). In this part the extended homogeneity range of the CoAl phase is a solid solution being in equilibrium with the liquid phase. Solidification of an alloy in this composition range starts with the formation of a (in comparison to the composition of the melt) Co-poor nucleus which on further cooling grows by formation of shells with continuously increasing Co content. For example, in the alloy Al–59.4Co the content of Co varies continuously from about 55 at.% in the core of the dark areas to about 70 at.% in the white boundary regions (Fig. 2b). In agreement with the phase diagram, the composition gradients become more pronounced with increasing Co content as can be concluded from the standard deviations of the EPMA values used for alloy composition analysis (2.4, 3.2, and 4.6 at.% for the alloys with 54.8, 56.4, and 59.4 at.% Co, respectively). As the concentration gradients are continuous, i.e., there is no jump which would indicate a phase boundary, and as DTA samples did not show any effect indicating a phase transformation, it can be concluded that the as-cast alloys remain in a metastable, single-phase CoAl state at low temperatures.

In accordance with the phase diagram, the two Al-rich alloys with 30.5 and 40.7 at.% Co show peritectic microstructures consisting of the primary phase CoAl surrounded by Al_5Co_2 (see Fig. 2c,d). In case of the Al-richest alloy Al–30.5Co, additionally some Al_3Co has formed. It is likely that this phase results from some remnant liquid phase which has not completely solidified during cooling on passing the peritectic temperature of Al_5Co_2 formation.

2.3. Melting temperatures of CoAl

All alloys were studied by DTA applying constant heating and cooling rates of 5 K/min between room and melting temperature in at least two cycles. The only effects visible on the DTA curves were those related to the melting process; no solid state reactions were observed in any of the alloys. Fig. 3 shows the high-temperature range of some representative heating curves. The solidus and liquidus temperatures of the non-stoichiometric intermetallic compounds were determined from the DTA curves by applying the following procedure:

Solidus temperatures were exclusively determined from heating curves where they correspond to the onset of the melting effect, i.e., the first deviation from the base line. However, as the exact position of this point often is difficult to determine (for example because DTA base lines in most cases are not linear but show some curvature), the intersection of the extrapolated base line (indicated by the dotted lines in Fig. 3) and a linear fit to the first part of the downward sloping section of the DTA signal is usually defined as the solidus temperature. This procedure, which follows the definition given by Höhne et al. [28,29], was also applied to analyse the present experiments.

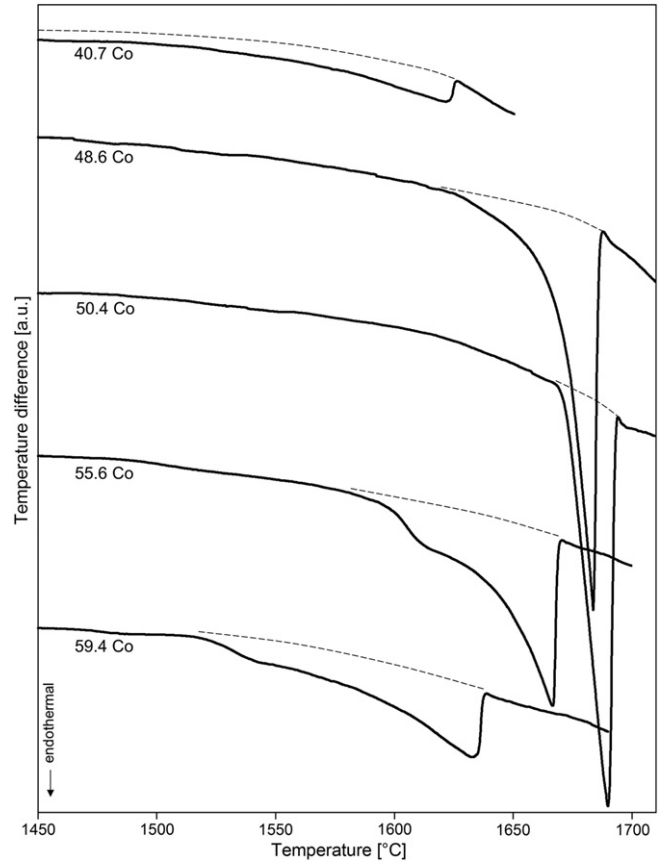

Fig. 3. High-temperature part of DTA heating curves (5 K/min) showing the melting range of representative Al–Co alloys.

The final liquidus temperatures were determined using both the heating and cooling curves as is described in the following: For an ideal, stoichiometric intermetallic compound, a heating curve shows a melting effect with the downward sloping part of the peak being completely linear. Exactly as in case of pure metals, the liquidus temperature then is identical to the solidus temperature and corresponds to the onset point on the heating curve as described above. In case of compositions deviating from the stoichiometry, the downward sloping part consists of two regions, a curved one which corresponds to the temperature interval of the two-phase field liquid + intermetallic phase and ends at the liquidus temperature, and a linear one starting at the liquidus temperature. That means the characteristic point defining the liquidus temperature is at the beginning of the linear part, which in some cases is difficult to determine. In case of alloys with a large temperature interval between solidus and liquidus (as, e.g., the alloys with 40.7, 55.6 and 59.4 at.% Co), the complete downwards sloping part of the melting effect is curved meaning that in these cases the characteristic point is near or at the peak temperature. On the other hand, in case of alloys with compositions near or at the stoichiometry, where the melting interval becomes very small (e.g. the alloy with 50.4 at.% Co in Fig. 3), the main part of the downwards sloping section is linear and the characteristic temperature is near to the solidus point. In all cases the peak temperature of the endothermic effect on the heating curve gives an upper limit for the liquidus.

A second value for the liquidus temperature is defined by the easily to determine onset point of solidification on the cooling curves.

Due to possible undercooling effects, this value gives a lower limit for the liquidus temperature. The final liquidus temperatures listed in Table 1 are average values of those derived from the heating curves (giving upper limits) and cooling curves (giving lower limits).

Fig. 4 shows the revised melting range of the phase diagram including the present results and all melting temperature data available from literature. The new value of 1673 °C for the stoichiometric melting point of CoAl obviously is significantly higher than those reported by Gwyer (1628 °C) [5] and assessed by McAlister (1640 °C) [13]. Above 60 at.% Co, the solidus and liquidus temperatures measured by Kimura et al. [20] fairly well lie on the extrapolated lines of the present solidus and liquidus data to the eutectic CoAl + (αCo) reaction at 1400 °C and 81.0 at.% Co (and a CoAl composition of 78.5 at.% Co) [13], whereas the values of Gwyer [5], especially at the higher temperatures, do not fit well.

The enthalpy of melting ΔH_{melt} of the single-phase CoAl alloys is plotted in Fig. 5 as a function of the Co content. Irrespective of the comparably low accuracy of the measured values, the data clearly show that ΔH_{melt} reduces to about half of its value when the Co content increases from the stoichiometric value to 60 at.%. In the Landolt-Börnstein database [30] (which uses the thermodynamic description from [14]), a melting enthalpy of 25510 J/mol (corresponding to 594 J/g provided that 'mol' here means 'mole of atoms') is reported for stoichiometric CoAl. This value very well agrees with the present result (590 J/g).

2.4. Al-rich phase boundary of B2 CoAl

Two specimens of the Al-rich two-phase alloy Al–40.7Co were heat-treated at 900 °C (3 weeks) and 1100 °C (1 week). The composition analyses by EPMA (see Table 2) confirm a low solubility of Al in the *B*2 CoAl phase – as was already indicated by the dotted line in the old phase diagram in Fig. 1 – and indicate a slight decrease of the Co content with increasing temperature. The literature value of 46.4 at.% Co at 1000 °C (extrapolated by McAlister [13] from the data of Ettenberg et al. [25]) is in the same range.

From the present data, a value of 46.0 at.% Co can be tentatively estimated for the CoAl composition at the peritectic reaction with the Al_5Co_2 phase (see section 2.5 below) in good agreement with the literature value of 46.3 at.% Co [13].

2.5. Al_5Co_2

As mentioned in section 2.2, the two Al-rich alloys with 30.5 and 40.7 at.% Co reveal the peritectic formation of the phase Al_5Co_2. DTA curves show a distinct peak at this peritectic reaction with onset temperatures of 1180 and 1181 °C, respectively. These values are in good agreement with literature data. In his first study of the Al–Co phase diagram, Gödecke [31] reported a measured value of 1180 °C. This temperature was also used by McAlister in his assessment [13], but was slightly changed to 1182 °C in a later publication of Gödecke and Ellner [9]. A higher value of 1188 °C was reported by Grushko et al. [11]. The reason for this increased value is not clear and it can only be speculated that the difference is related to the applied heating rates. Similar as in the present experiments, Gödecke used DTA heating rates in the range 2–10 K/min, whereas Grushko et al. measured their samples with a comparably high heating rate of 20 K/min.

Fig. 6 shows a DTA heating curve of an as-cast sample of the alloy Al–30.5Co, which contained some metastably retained Al_3Co as already mentioned in section 2.2 and visible in the microstructure shown in Fig. 2c. The small peak with an onset temperature of 1153 °C belongs to the peritectic decomposition reaction of Al_3Co. The same value for this reaction was also measured by Gödecke and Ellner [9] and once again the respective value reported by Grushko et al. [11] is a little bit higher (1158 °C). After holding for 50 h at 1100 °C, the small peak at 1153 °C has completely vanished proving that the small amount of Al_3Co was only metastably retained in the as cast sample.

The heat-treated samples of alloy Al–40.7Co reveal two-phase microstructures consisting of the phases CoAl + Al_5Co_2; an example is shown in Fig. 2e. The values measured by EPMA for the equilibrium composition of the Al_5Co_2 phase at 900 and 1100 °C are given in Table 2. As the theoretical composition of stoichiometric Al_5Co_2 is

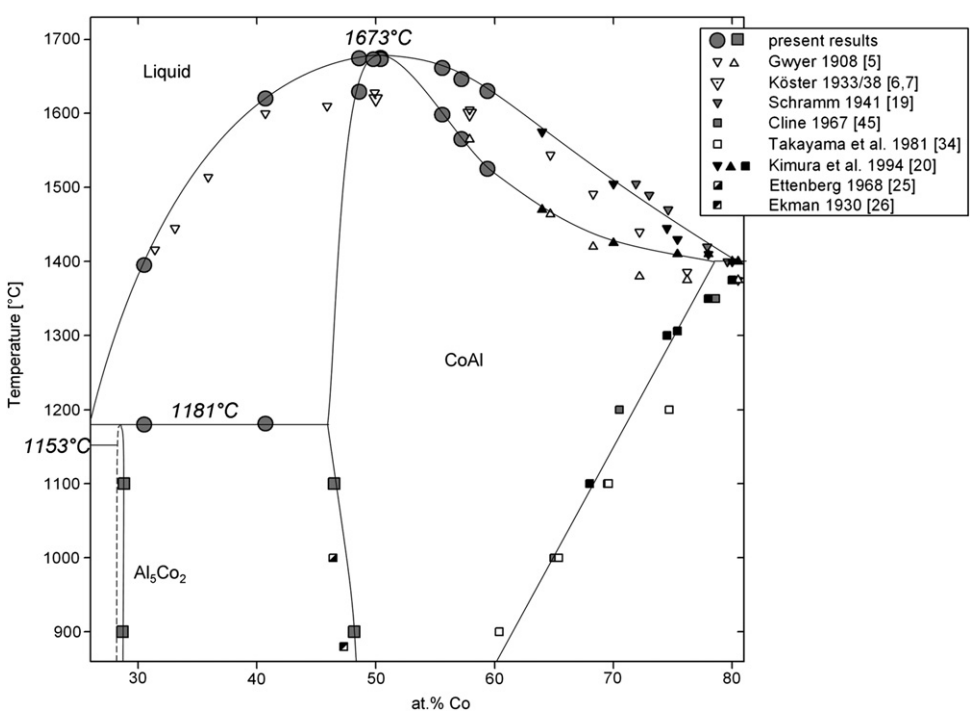

Fig. 4. Experimentally determined part of the Al–Co phase diagram showing the melting range of the CoAl phase (upwards and downwards pointing triangles represent literature values from [5–7,19,20] for solidus and liquidus temperatures, respectively; squares are data for phase boundaries in the solid state [20,25,26,34,45]).

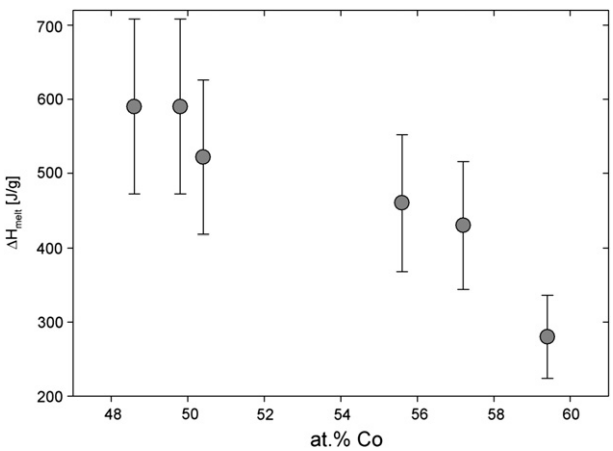

Fig. 5. Enthalpy change ΔH_{melt} on melting single-phase CoAl as obtained from DTA.

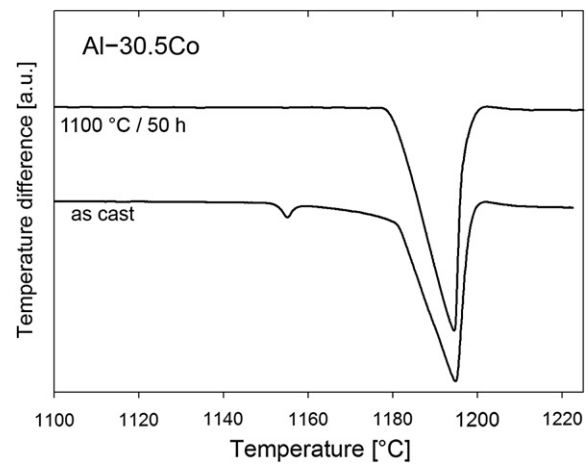

Fig. 6. DTA heating curve (5 K/min) of the alloy Al–30.5Co; the curve for the as-cast sample shows the two peritectic decomposition reactions of Al_5Co_2 at 1180 °C and of some metastably in the as-cast state retained Al_3Co at 1153 °C, the upper curve shows that the effect of the Al_3Co decomposition vanishes after proper annealing.

28.6 at.% Co, these values indicate that the phase has a very narrow homogeneity range.

3. Thermodynamic modelling

In order to take into account the new experimental results, the thermodynamic description of the complete Al–Co system was updated. Starting point for the re-assessment was the work of Dupin and Ansara [14].

3.1. Choice of experimental data

The experimental data reported in the literature were already critically reviewed by McAlister [13] and Dupin and Ansara [14]. Here, we only briefly present the experimental data which were used in the present assessment.

In the Al-rich part, the present experimental data on the peritectic reactions L + AlCo → Al_5Co_2 (occurring at 1181 °C) and L + Al_5Co_2 → Al_3Co (occurring at 1153 °C) were considered in the assessment work (see also section 2.5 above). Gödecke and Ellner [9] and Grushko et al. [11] report on the existence of three different intermetallic phases in the composition range of the phase $Al_{13}Co_4$. However, as the results regarding composition and temperature ranges of these phases are quite contentious, a single phase $Al_{13}Co_4$ as also shown in McAlister's review [13] has been used for the present modelling. For the liquidus temperatures in this composition range, data reported by Gödecke and Ellner [9] were considered. At the very Al-rich end of the composition range between 0 and ~4 at.% Co, a high weight was put on the liquidus data reported by Fink and Freche [32]. All other relevant data on the Al-rich end of the phase diagram were taken from McAlister [13].

For the CoAl phase, mainly the present experimental results were adopted. For the Co-rich part of the phase diagram, such as the solidus and liquidus of (Co) and the boundary between CoAl and (Co) phase, literature data from Kimura et al. (DTA and EDS) [20], Dutchak and Chekh (high temperature XRD) [33], Takayama et al. (EPMA) [34] and Gómez-Acebo et al. (EPMA) [35] were used in the present modelling.

Table 2
Phase compositions of the heat-treated alloy Al–40.7Co as determined by EPMA (the given values are the average values of 12–16 measured points and the respective standard deviations).

Heat treatment temperature and time	Phase composition (at.% Co)	
	CoAl	Al_5Co_2
900 °C/3 weeks	48.2 ± 0.2	28.7 ± 0.2
1100 °C/1 week	46.5 ± 0.5	28.8 ± 0.3

Data for enthalpies of formation, which were determined by Henig et al. [36] applying calorimetric measurements, were also taken into account to model the CoAl phase. The Al chemical potential in the Co-rich liquid phase at 1600 and 1677 °C was calculated on the basis of activity data a_{Al} reported by Vachet et al. [37] (for 1600 °C) and Levin and Gel'd [38] (for 1677 °C), and the results were used in the present calculations. From the melting enthalpy data for CoAl, which were measured in this work, only the value for the alloy with 49.8% Co was used since it is closest to the congruent melting point. However, because of its high estimated error of about 20% and the slight deviation from the ideal stoichiometric composition, only a low weight was given to this value.

The Curie temperatures of the (αCo) solid solution reported in [7,20] were used to optimize the magnetic binary interaction parameter of the (αCo) phase.

3.2. Thermodynamic modelling of the phases

For the present calculations, all phases of the binary system, i.e. the liquid phase, fcc_A1 (Al, αCo), bcc_A2 and bcc_B2 (CoAl), and Al_9Co_2, $Al_{13}Co_4$, Al_3Co and Al_5Co_2 were modelled and optimized. The *A*2 phase is not stable in this simple system but there are many systems, binary like Al–Fe or ternary like Al–Co–Cr based on the system here under consideration, where this phase shows a second order relation with the *B*2 phase in some composition/temperature ranges, see e.g [46,47]. The formalism used here to describe the *B*2 phase allows to describe such relations and – even though the *B*2 structure from a crystallographic point of view is not bcc but simple cubic – the designation 'bcc_B2' is used in order to emphasize this behaviour of the description and to remind the reader that its description has a disordered contribution from the metastable bcc_A2 phase.

The total molar Gibbs energy of a phase θ can be expressed in a general form (see, e.g., [39],) as

$$G_m^\theta = {}^0G_m^\theta + {}^{id}G_m^\theta + {}^{mgn}G_m^\theta + {}^{ex}G_m^\theta, \quad (1)$$

where ${}^0G_m^\theta$ represents the Gibbs energy of an unreacted mixture of the constituents of the phase, ${}^{id}G_m^\theta$ is the configuration contribution standing for the ideal entropy of mixing, ${}^{ex}G_m^\theta$ is the excess Gibbs

Table 3
Updated thermodynamic parameters for the Al–Co system.

Liquid
(Al,Co)
$^0G^{Liq}_{Al} - {}^0H^{fcc_A1}_{Al} = GALLIQ$
$^0G^{Liq}_{Co} - {}^0H^{hcp_A3}_{Co} = GCOLIQ$
$^0L^{Liq}_{Al,Co} = -150510 + 33.729 \cdot T$
$^1L^{Liq}_{Al,Co} = -54090 + 26.8 \cdot T$
$^2L^{Liq}_{Al,Co} = 65430 - 22.4 \cdot T$

CoAl
bcc_A2: $(Al,Co,Va)_1$
$^0G^{bcc_A2}_{Al} - {}^0H^{fcc_A1}_{Al} = GALBCC$
$^0G^{bcc_A2}_{Co} - {}^0H^{hcp_A3}_{Co} = GCOBCC$
$^0T_{C\,Co}^{bcc_A2} = 1450$
$^0\beta_{0\,Co}^{bcc_A2} = 1.35$
$^0G^{bcc_A2}_{Va} = 30 \cdot T$
$^0G^{bcc_A2}_{Al,Va} = 46912$
$^0G^{bcc_A2}_{Al,Co} = B2AlCo + LB2AlCo$
$^0L^{bcc_A2}_{Co,Va} = 126184$
$B2AlCo = -138247 + 29.92 \cdot T$
$LB2AlCo = 43040 - 27.4 \cdot T$

bcc_B2: $(Al,Co,Va)_{0.5}$ $(Al,Co,Va)_{0.5}$
$^0G^{bcc_B2}_{Al:Al} = 0$
$^0G^{bcc_B2}_{Co:Co} = 0$
$^0G^{bcc_B2}_{Va:Va} = 0$
$^0G^{bcc_B2}_{Al:Co} = {}^0G^{bcc_B2}_{Co:Al} = 4 \cdot UBAlCo$
$^0T_{C\,Al:Co}^{bcc_B2} = {}^0T_{C\,Co:Al}^{bcc_B2} = -1400$
$^0G^{bcc_B2}_{Al:Va} = {}^0G^{bcc_B2}_{Va:Al} = 0$
$^0G^{bcc_B2}_{Co:Va} = {}^0G^{bcc_B2}_{Va:Co} = 0$
$^0L^{bcc_B2}_{Co,Va:Al} = {}^0L^{bcc_B2}_{Al:Co,Va} = -56770 - 18 \cdot T$
$UBAlCo = 0.125 \cdot B2AlCo - 0.125 \cdot LB2AlCo$

(Al) and (αCo)
fcc_A1: (Al,Co)
$^0G^{fcc_A1}_{Al} - {}^0H^{fcc_A1}_{Al} = GHSERAL$
$^0G^{fcc_A1}_{Co} - {}^0H^{hcp_A3}_{Co} = GCOFCC$
$^0T_{C\,Co}^{fcc_A1} = 1396$
$^0\beta_{0\,Co}^{fcc_A1} = 1.35$
$^0T_{C\,Al,Co}^{fcc_A1} = -1500$
$^1T_{C\,Al,Co}^{fcc_A1} = 650$
$^0\beta_{0\,Al,Co}^{fcc_A1} = 10$
$^0L^{fcc_A1}_{Al,Co} = -124200 + 17.24 \cdot T$
$^1L^{fcc_A1}_{Al,Co} = 0$
$^2L^{fcc_A1}_{Al,Co} = 28740$

(εCo)
hcp_A3: (Al,Co)
$^0G^{hcp_A3}_{Al} - {}^0H^{fcc_A1}_{Al} = GALHCP$
$^0G^{hcp_A3}_{Co} - {}^0H^{hcp_A3}_{Co} = GHSERCO$
$^0T_{C\,Co}^{hcp_A3} = 1396$
$^0\beta_{0\,Co}^{hcp_A3} = 1.35$

Al_5Co_2
$^0G_{Al_5Co_2} - 5 \cdot {}^0H^{fcc_A1}_{Al} - 2 \cdot {}^0H^{hcp_A3}_{Co} = -329990 + 73.45 \cdot T + 5 \cdot GHSERAL + 2 \cdot GHSERCO$

Al_3Co
$^0G_{Al_3Co} - 3 \cdot {}^0H^{fcc_A1}_{Al} - {}^0H^{hcp_A3}_{Co} = -164946 + 32.725 \cdot T + 3 \cdot GHSERAL + GHSERCO$

$Al_{13}Co_4$
$^0G_{Al_{13}Co_4} - 13 \cdot {}^0H^{fcc_A1}_{Al} - 4 \cdot {}^0H^{hcp_A3}_{Co} = -659712 + 125 \cdot T + 13 \cdot GHSERAL + 4 \cdot GHSERCO$

Al_9Co_2
$^0G_{Al_9Co_2} - 9 \cdot {}^0H^{fcc_A1}_{Al} - 2 \cdot {}^0H^{hcp_A3}_{Co} = -329700 + 53.14 \cdot T + 9 \cdot GHSERAL + 2 \cdot GHSERCO$

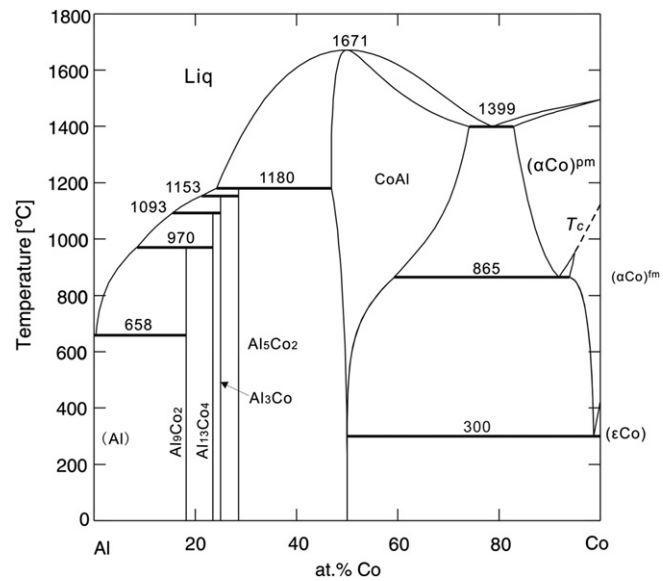

Fig. 7. Calculated Al–Co phase diagram.

energy, and $^{mgn}G^{\theta}_m$ represents the magnetic contribution to the Gibbs energy.

For the regular solution phases, such as liquid, fcc_A1 and bcc_A2, $^0G^{\theta}_m$ and $^{id}G^{\theta}_m$ can be expressed as

$$^0G = \sum_i x_i {}^0G(T) \text{ and } {}^{id}G^{\theta}_m = RT\sum_i x_i \ln(x_i), \tag{2}$$

where x_i is the mole fraction of the component i and $^0G_i(T)$is its molar Gibbs energy. The factor R in the configuration term is the gas constant and T is the absolute temperature. The excess Gibbs energy $^{ex}G^{\theta}_m$ is described by a Redlich-Kister series:

$$^{ex}G^{\theta}_m = x_{Al}x_{Co}\sum_{\nu} {}^{\nu}L_{Al,Co}(x_{Al} - x_{Co})^{\nu} \tag{3}$$

The optimized interactive parameter $^{\nu}L_{Al,Co}$ can be temperature-dependent.

The magnetic contribution term is expressed as

$$G^{mag} = RT\ln(\beta_0 + 1)f(\tau), \tag{4}$$

where β_0 is the average Bohr magneton number per atom and $f(\tau)$ is the integral of a function fitted to the magnetic contribution to the

Table 4
Comparison of calculated and experimental invariant reaction temperatures in the Al–Co binary system ($(\alpha Co)^{pm}$ and $(\alpha Co)^{fm}$ refer to the para- and ferromagnetic state of the Co solid solution).

Invariant reaction	Reaction type	T_{exp} (°C)	Reference	T_{calc} (°C)
Liq ↔ CoAl	Congruent	1673	This work	1671
Liq ↔ CoAl + $(\alpha Co)^{pm}$	Peritectic	1400	[13]	1399
Liq + CoAl ↔ Al_5Co_2	Peritectic	1181	This work	1180
		1182	[9]	
		1188	[11]	
		1180	[13]	
Liq + Al_5Co_2 ↔ Al_3Co	Peritectic	1153	This work	1153
		1153	[9]	
		1158	[11]	
		1135	[13]	
Liq + Al_3Co ↔ $Al_{13}Co_4$	Peritectic	1093	[13]	1093
Liq + $Al_{13}Co_4$ ↔ Al_9Co_2	Peritectic	970	[13]	970
$(\alpha Co)^{pm}$ ↔ CoAl + $(\alpha Co)^{fm}$	Eutectoid	–	–	865
Liq ↔ Al_9Co_2 + (Al)	Eutectic	657	[13]	658
CoAl + $(\alpha Co)^{fm}$ ↔ (εCo)	Eutectoid	~300	[13]	300

heat capacity with $\tau = T/T_C$, and T_C being the Curie temperature [40]. The composition dependence of the Curie temperature and the Bohr magneton number is described in the same way as for the non-magnetic excess Gibbs energy:

$$T_C^\theta = x_{Co}T_{CCo}^\theta + x_{Al}x_{Co}\sum_\nu {}^\nu T(x_{Al} - x_{Co})^\nu \quad (5)$$

$$\beta_0^\theta = x_{Co}\beta_{0Co}^\theta + x_{Al}x_{Co}{}^0\beta_{Al,Co}^\theta \quad (6)$$

The four Al-rich intermetallic phases Al_9Co_2, $Al_{13}Co_4$, Al_3Co and Al_5Co_2 were modelled as stoichiometric phases in the same way as by Dupin and Ansara [14].

The *B*2-ordered CoAl phase was modelled with a two-sublattice model $(Al,Co,Va)_{0.5}(Al,Co,Va)_{0.5}$. Although disordered *A*2 has been observed as a metastable phase in the system [21–23], a single function to describe the Gibbs energy of ordered *B*2 and disordered *A*2 phase together was used in this work. The single function is defined by the following equations:

$$G_m^{total} = G_m^{dis}\left(X\right) + \Delta G_m^{ord} \text{ with} \quad (7)$$

$$\Delta G_m^{ord} = G^\bullet\left(y_i^{(s)}\right) - G^\bullet\left(y_i^{(s)} = x_i\right) \quad (8)$$

The first term $G_m^{dis}(x)$ in Eq (7) is the Gibbs energy of the disordered *A*2 phase and can be calculated by the solution model described above. The second term ΔG_m^{ord} gives the long-range ordering contribution and depends on the site fraction $y_i^{(s)}$ (the fraction of the constituent *i* in sublattice *s*). It will always be zero if the phase is disordered, since in the disordered state $y_i' = y_i'' = x_i$ and, therefore, the first term $G^\bullet(y_i^{(s)})$ in Eq (8) is equal to the second term $G^\bullet(y_i^{(s)} = x_i)$.

The crystallographic equivalence of the two sublattices of the *B*2 structure means that exchanging the occupation of the two sublattices should result in an identical energy, i.e.: ${}^0G_{A:B} = {}^0G_{B:A}$ and ${}^\nu L_{A,B:i} = {}^\nu L_{i:A,B}$.

A more detailed introduction to the single function description of the disordered *A*2 and ordered *B*2 phase is given in [41].

3.3. Calculated results and comparison with experiments

The optimisation of the thermodynamic parameters of all phases was performed with the PARROT part of the Thermo-Calc software package. All parameters for all phases have been reassessed in the present work and are given in Table 3. The calculated phase diagram is shown in Fig. 7. The invariant reaction temperatures are listed in Table 4.

Fig. 8a shows the Al-rich part of the phase diagram. The four intermediate phases, which all were modelled as stoichiometric compounds, form in peritectic reactions in agreement with the experimental information. All calculated and experimentally determined invariant reaction temperatures agree within 2 K (see Table 4). The liquidus line in the Al-rich corner well fits to the data of Fink and Freche [32] and Gödecke and Ellner [9].

The CoAl phase region in the centre of the phase diagram is shown in Fig. 8b, All new experimental data of the present work were accounted for and are well fitted. The calculated CoAl phase has a congruent melting point at $x_{Co} = 0.5003$ and 1671 °C.

The Co-rich part of the phase diagram is shown in Fig. 8c. The agreement of experimental and theoretical values for the Al solubility in the fcc phase is good. The calculated phase boundary of CoAl well fits to most of the experimental data, and the maximum Co content in CoAl was calculated to be $x_{Co} = 0.741$ at 1399 °C. The calculated liquidus line not only has a good agreement with the new experimental data, but also describes very well Kimura's experimental data [20]. By adjusting the parameters ${}^0T_{CAl,Co}^\theta$ and ${}^1T_{C\ Al,Co}^\theta$ in the T_C term, a good match between calculated and experimental ([7,20,42]) Curie temperatures of (αCo) is achieved.

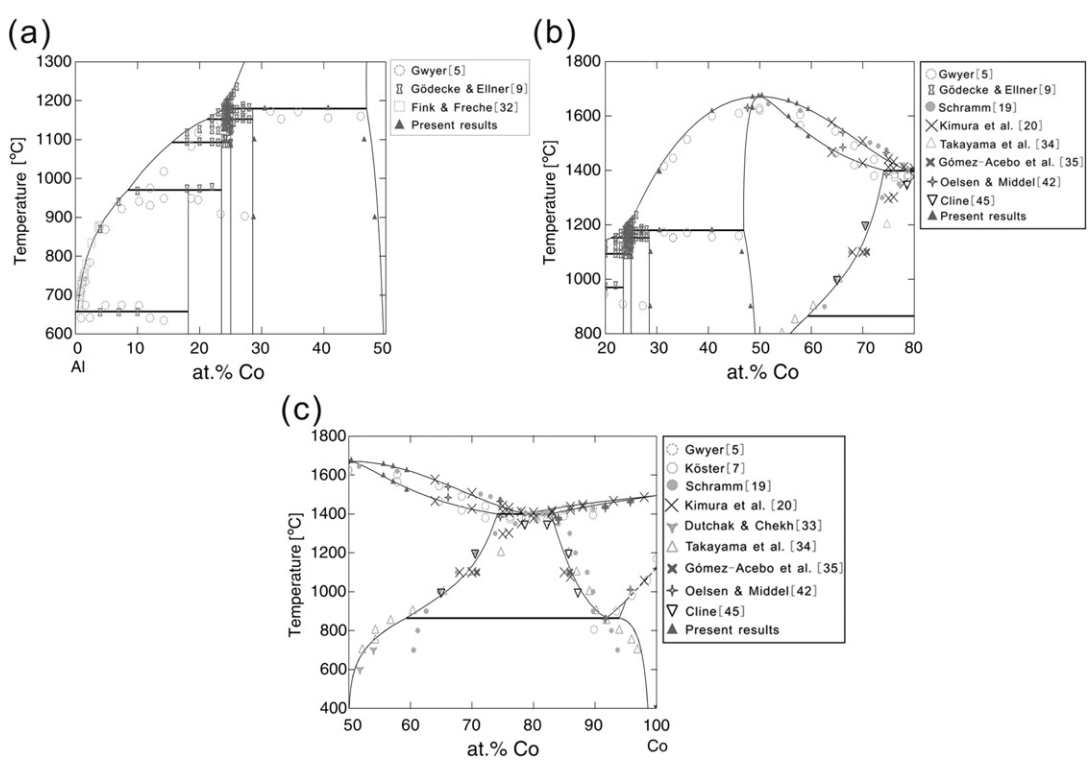

Fig. 8. Calculated partial phase diagram in comparison with experimental data from the literature [5,7,9,19,20,32–35,42,45]: (a) Al-rich side, (b) Central part, (c) Co-rich side.

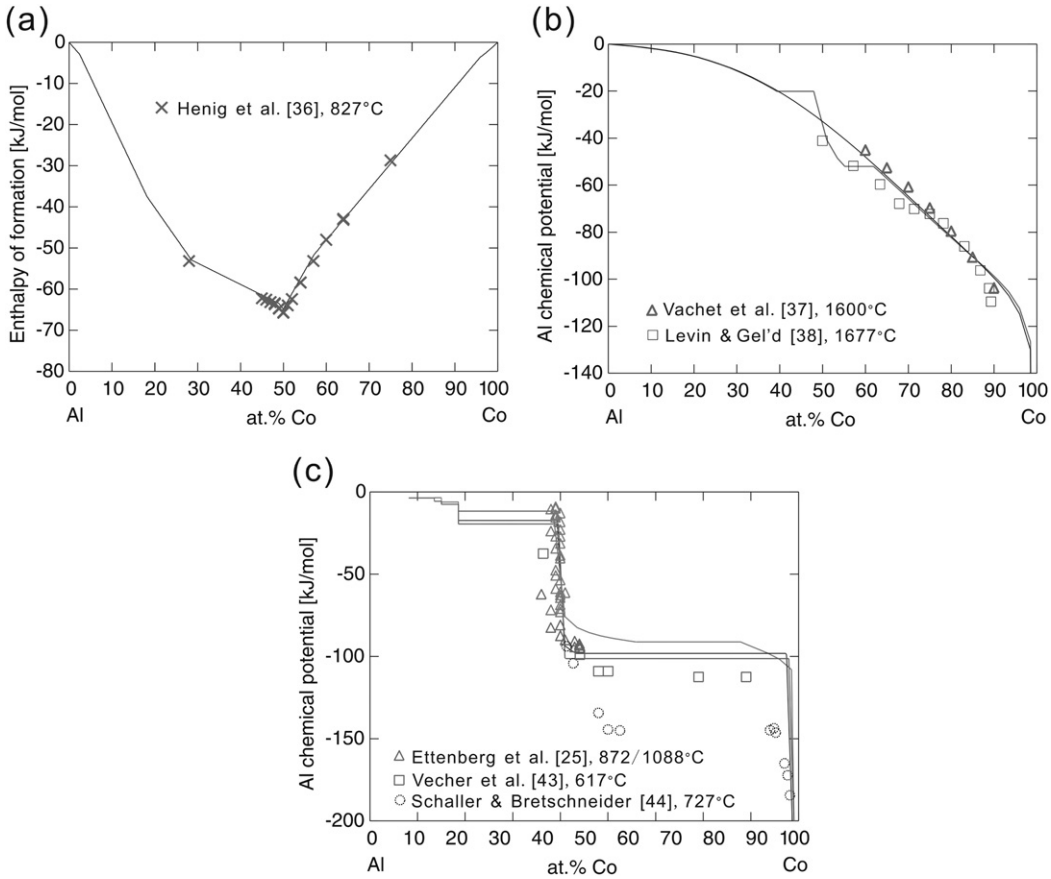

Fig. 9. (a) Calculated enthalpies of formation (solid line) compared with the calorimetric data of Henig et al. [36], (b) Calculated Al chemical potential of the liquid compared with literature data [37,38], and (c) calculated Al chemical potential of solid compared with literature data [25,43,44].

However, this leads at the same time to the prediction of the occurrence of a small miscibility gap between ferro- and paramagnetic (αCo) (see Fig. 8c), which up to now has not been confirmed experimentally.

Fig. 9a shows the calculated enthalpies of formation for the Al–Co system revealing excellent agreement with results from calorimetric measurements [36]. The calculated Al chemical potential in the liquid phase is shown in Fig. 9b, the agreement with the experimental data is good. Fig. 9c presents the calculated Al chemical potential in the solid state together with measured data from [25,43,44]. These experimental data are only used for comparison and were not included in the assessment work.

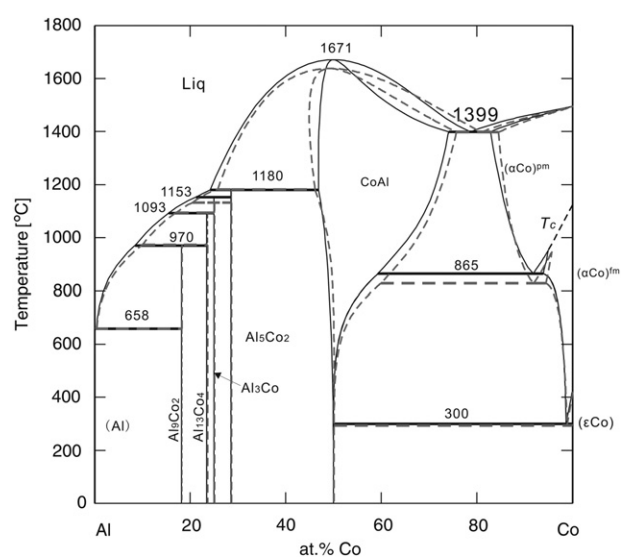

Fig. 10. Comparison of the Al–Co phase diagram calculated in the present work (solid line) with the version calculated in Dupin and Ansara's work (dashed line) [14].

Nevertheless, they are well fitted by the present thermodynamic description.

In Fig. 10, a comparison between the present calculations and those reported by Dupin and Ansara [14] is given. Because of the new experimental data for the CoAl phase, especially the central part of the phase diagram is described much better now.

4. Summary and conclusions

The central part of the Al–Co phase diagram was investigated experimentally, and on the basis of these new results and further reliable literature data, the whole binary Al–Co system has been reassessed. The following summarizing conclusions can be drawn:

- The melting temperature of stoichiometric CoAl was found to be 1673 ± 3 °C being 33 °C higher than the hitherto accepted and used value.
- The enthalpy of melting of stoichiometric CoAl is 590 J/g corresponding to 25.3 kJ/(mole of atoms). It decreases strongly as a function of Co content reaching about half of its value for single-phase CoAl with 60 at.% Co.
- Whereas as-cast Al–Co alloys near the stoichiometric CoAl composition have homogeneous microstructures, Co-rich single-phase CoAl alloys reveal strong fluctuations in composition with a grain-like appearing microstructure consisting of Al-enriched cores and Co-enriched boundary zones.
- The homogeneity range of CoAl is very unsymmetrical with respect to the stoichiometric composition. Whereas the solubility for Co is known from the literature to reach a maximum value of nearly 30 at.% (at 1400 °C), the present results show that the solubility for Al is comparably low amounting to 1.8 at.% at 900 °C and 3.5 at.% at 1100 °C. For the maximum solubility, which is reached at the temperature of the peritectic reaction, a value of 4 at.% is estimated.
- For the solubility of Co in the peritectically forming Al-rich phase Al_5Co_2, very low values of 0.1–0.2 at.% were found at 900 and 1100 °C indicating a very narrow homogeneity range for this phase. For the peritectic reactions $L + CoAl \leftrightarrow Al_5Co_2$ and $L + Al_5Co_2 \leftrightarrow Al_3Co$, temperatures of 1181 and 1153 °C were measured in good agreement with literature values.
- A re-assessment of the Al–Co system was carried out by the Calphad method to take into account the new experimental data. Relevant literature data and the present experimental results are well described by the optimized thermodynamic parameters. All calculated invariant reaction temperatures including the congruent melting temperature of CoAl agree within 2 K with the experimental data.

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